Introduction to Mathematics and Optimization

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The references in the suggested solutions to this examination paper are with respect to the PDF edition *Introduction to Mathematics and Optimization* (Compressed static version (15.11.2021) of interactive book).

PROBLEM 1

Let $f : \mathbb{R}^3 \to \mathbb{R}$ be the function given by

$$f(x, y, z) = x^2 + y^2 + (z - 1)^2.$$

Furthermore, let

$$\begin{aligned} g_1(x, y, z) &= z - x \\ g_2(x, y, z) &= z + x \\ g_3(x, y, z) &= z - y \\ g_4(x, y, z) &= z + y \\ g_5(x, y, z) &= -1 - z \end{aligned}$$

be functions from $\mathbb{R}^3$ to $\mathbb{R}$ and define

$$C = \{v \in \mathbb{R}^3 \mid g_1(v) \leq 0, \quad g_2(v) \leq 0, \quad g_3(v) \leq 0, \quad g_4(v) \leq 0, \quad g_5(v) \leq 0\}$$

for $v = (x, y, z) \in \mathbb{R}^3$. In this problem we focus on the optimization problem

$$\begin{array}{c} \min f(v) \\ v \in C. \end{array} \tag{1}$$

(a) Show that (1) is a convex optimization problem.

ANSWER:

The Hessian matrix of f is $\begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$. This matrix is positive semidefinite, since it is a diagonal matrix with non-negative entries (Exercise 8.26). Hence f is a convex function by Theorem 8.21. The subset C is an intersection of sets of the form $\{v \in \mathbb{R}^d \mid f(v) \leq a\}$, where $f : \mathbb{R}^d \to R$ is convex (Lemma 4.26). In the present case, each of these functions has a Hessian matrix with zero entries (again concluding by Theorem 8.21). Thus C is convex, since it is an intersection of five convex subsets (Exercise 4.22). Therefore (1) is a convex optimization problem.

(b) The condition $(x, y, z) \in C$ means that

$$z \leq x, \qquad z \leq -x, \qquad z \leq y, \qquad z \leq -y, \qquad z \geq -1. \tag{2}$$

Show that from $z \leq x$ and $z \leq -x$ one can conclude that $z \leq 0$. Explain likewise why one can conclude that $-1 \leq x \leq 1$ and $-1 \leq y \leq 1$ based on the inequalities in (2).

ANSWER:

We have $z \le x$ and $x \le -z$, hence $z \le -z$, which implies $2z \le 0$ and thus $z \le 0$. For y we obtain that $-1 \le z$ and $z \le y$ implies $-1 \le y$. Similarly, $y \le -z \le 1$, which implies $y \le 1$. The argument for the inequalities with respect to x is the same.

(c) Show that C is compact. Explain why this implies that the optimization problem (1) can be solved.

ANSWER:

A set being compact means that it is closed and bounded according to Definition 5.75. C is closed, since it is an intersection of five closed sets of the form $\{v \in \mathbb{R}^d \mid f(v) \le a\} = f^{-1}((-\infty, a])$, where $f : \mathbb{R}^d \to \mathbb{R}$ is a continuous function (Proposition 5.72). The continuity of these functions follows from Lemma 5.62 and Proposition 5.64. Furthermore, $(-\infty, a]$ is a closed subset of $\mathbb{R}$ by Proposition 5.51. That C is bounded follows from part (b): $(x, y, z) \in C \implies x^2 + y^2 + z^2 \le 1^2 + 1^2 + 1^2 = 3$, and C is bounded according to §5.2.3. The optimization problem has a solution by Theorem 5.76, since C is compact and $f(x, y, z) = x^2 + y^2 + (z-1)^2$ is continuous, as it is convex and defined on an open set (Theorem 6.6).

(d) Is a solution to the optimization problem unique? If so, why?

ANSWER:

Yes, a solution exists according to the previous part, and it must be unique by Theorem 6.12, since f is a strictly convex function, as its Hessian matrix is a diagonal matrix with positive numbers on the diagonal (cf. Theorem 8.21 and Exercise 8.26).

(e) Write down the KKT conditions corresponding to (1) in the variables $x, y, z, \lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5$ and explain from these why $(-1, -1, -1)$ cannot be an optimal solution to (1).

ANSWER:

According to (9.24), the KKT conditions are

$$\begin{aligned}
&\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5 \ge 0 \\
&\lambda_1(z - x)) = 0 \\
&\lambda_2(z + x) = 0 \\
&\lambda_3(z - y) = 0 \\
&\lambda_4(z + y) = 0 \\
&\lambda_5(-1 - z) = 0 \\
&2x - \lambda_1 + \lambda_2 = 0 \\
&2y - \lambda_3 + \lambda_4 = 0 \\
&2(z - 1) + \lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 - \lambda_5 = 0
\end{aligned}$$

For $v = (x, y, z) = (0, 0, -\frac{1}{2})$ we obtain strict inequalities $g_1(v), \dots, g_5(v) < 0$. By Definition 9.29, the optimization problem is strictly feasible. The point $w = (-1, -1, -1) \in C$ gives rise to $\lambda_2 = \lambda_4 = 0$ and hence the equation $-2 - \lambda_1 = 0 \implies \lambda_1 = -2$, which contradicts $\lambda_1 \ge 0$. Theorem 9.30 (i) then states that w cannot be an optimal solution.

(f) Show that $(0, 0, 0)$ is an optimal solution to (1). Do the KKT conditions corresponding to (1) have a unique solution?

ANSWER:

We substitute $(0, 0, 0)$ into the KKT conditions and obtain $\lambda_5 = 0$ and

$$\begin{aligned}
\lambda_1 &= \lambda_2 \\
\lambda_3 &= \lambda_4 \\
-2 + 2\lambda_1 + 2\lambda_3 &= 0
\end{aligned}$$

We can thus choose arbitrary $\lambda_1, \lambda_3 \ge 0$ with $\lambda_1 + \lambda_3 = 1, \lambda_2 = \lambda_1$, and $\lambda_4 = \lambda_3$. These will solve the KKT conditions for $(x, y, z) = (0, 0, 0)$. Theorem 9.30 (ii) then gives that $(0, 0, 0)$ is an optimal solution. At the same time, it is evident that a solution to the KKT conditions is not unique.

(g) Give a geometric interpretation of the optimization problem (1).

ANSWER:

The function $f(x, y, z)$ expresses the square of the distance from the point $(0, 0, 1)$ to (x, y, z). An optimal solution to (1) is thus a point in C with minimal distance to $(0, 0, 1)$. One can sketch C and observe that it is a pyramid with apex at $(0, 0, 0)$.

PROBLEM 2

Let A denote the matrix

$$\begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$$

and $f : \mathbb{R}^2 \to \mathbb{R}$ the function given by

$$f(x, y) = x^2 + xy.$$

(a) Find an invertible matrix B such that

$$B^\intercal A B = \begin{pmatrix} 2 & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}.$$

ANSWER:

The matrix B corresponds to multiplying the first column by $-\frac{1}{2}$ and adding to the second column. This is given by

$$B = \begin{pmatrix} 1 & -\frac{1}{2} \\ 0 & 1 \end{pmatrix}$$

cf. Exercise 3.42 or Theorem 8.27 (the proof thereof). One can verify that B is invertible with inverse

$$B^{-1} = \begin{pmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{pmatrix}.$$

(b) Show that A is not positive semidefinite.

ANSWER:

This follows from Exercise 8.26 and Proposition 3.41.

(c) Show that f is not a convex function.

ANSWER:

The Hessian matrix of f is precisely A. We have shown that this is not positive semidefinite. The conclusion follows from Theorem 8.21.

(d) Find two vectors $u, v \in \mathbb{R}^2$ such that

$$u^\intercal A u > 0 \qquad \text{and}$$
$$v^\intercal A v < 0$$

possibly using the matrix B from (a).

ANSWER:

For

$$D = B^\intercal A B$$

we see that $e_1^\intercal D e_1 = 2 > 0$ and $e_2^\intercal D e_2 = -\frac{1}{2} < 0$, where $e_1 = (1, 0)^\intercal$ and $e_2 = (0, 1)^\intercal$. Hence we can use $u = Be_1 = (1, 0)^\intercal$ and $v = Be_2 = (-\frac{1}{2}, 1)^\intercal$.

(e) Find the critical points of f and determine their types (local minimum/maximum, saddle point, etc.).

ANSWER:

The critical points are the solution set of the equations

$$\frac{\partial f}{\partial x} = 2x + y = 0$$
$$\frac{\partial f}{\partial y} = x = 0$$

Hence f has only the critical point $(0, 0)$. The Hessian matrix at this point is precisely A, and it follows that it is indefinite from part (d) above. By Theorem 8.9 (iii), this unique critical point is a saddle point.

PROBLEM 3

Let

$$f(x) = a_2x^2 + a_1x + a_0 \tag{3}$$

denote a function $f : \mathbb{R} \to \mathbb{R}$ given by the real numbers a_2, a_1, a_0. For $a_2 > 0$, this is a familiar parabola opening upward.

(a) Compute $f''(x)$. What must a_2 satisfy for $f(x)$ to be a convex function? What must a_2 satisfy for $f(x)$ to be strictly convex?

ANSWER:

Since $f''(x) = 2a_2$, we have $a_2 \geq 0$ if and only if f is convex by Corollary 6.47. If $a_2 = 0$, then $f(x) = a_1x + a_0$ and hence not strictly convex. If $a_2 > 0$, then $f''(x) > 0$ and hence strictly convex, again by Corollary 6.47.

(b) Show that if the graph of the function f defined in (3) passes through the points $(0,0), (1,1), (2,3)$, and $(3,5)$, then the system of equations

$$\begin{aligned} a_0 &= 0 \\ a_2 + a_1 + a_0 &= 1 \\ 4a_2 + 2a_1 + a_0 &= 3 \\ 9a_2 + 3a_1 + a_0 &= 5 \end{aligned}$$

has a solution in a_2, a_1, and a_0. Show that this is impossible.

ANSWER:

The equations are a reformulation of $f(0) = 0, f(1) = 1, f(2) = 3$, and $f(3) = 5$, which express that the graph of f passes through the points. Since $a_0 = 0$, we must show that

$$\begin{aligned} a_2 + a_1 &= 1 \\ 4a_2 + 2a_1 &= 3 \\ 9a_2 + 3a_1 &= 5 \end{aligned}$$

has no solution. If we multiply the first equation by -2 and add to the second equation, we obtain $2a_2 = 1$ and hence $a_1 = a_2 = \frac{1}{2}$. This contradicts the last equation:

$$9 \cdot \frac{1}{2} + 3 \cdot \frac{1}{2} = 6 \neq 5.$$

(c) Find a best-fit parabola (in the least squares sense) through the points in part (b).

ANSWER:

We proceed from Theorem 5.19 with

$$x = \begin{pmatrix} a_2 \\ a_1 \\ a_0 \end{pmatrix}, \qquad A = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 1 & 1 \\ 4 & 2 & 1 \\ 9 & 3 & 1 \end{pmatrix} \qquad \text{and} \qquad b = \begin{pmatrix} 0 \\ 1 \\ 3 \\ 5 \end{pmatrix}$$

according to the unsolvable system of equations above. The coefficients of the best-fit parabola are given by the solution to

$$(A^\intercal A)v = A^\intercal b. \tag{4}$$

Computing, we obtain

$$A^\intercal A = \begin{pmatrix} 98 & 36 & 14 \\ 36 & 14 & 6 \\ 14 & 6 & 4 \end{pmatrix} \qquad \text{and} \qquad A^\intercal b = \begin{pmatrix} 58 \\ 22 \\ 9 \end{pmatrix}$$

Solving the system, we obtain

$$a_2 = \frac{1}{4}, \qquad a_1 = \frac{19}{20} \qquad \text{and} \qquad a_0 = -\frac{1}{20}.$$

(d) Find a number y such that there exists a parabola passing through the points $(0, 0)$, $(1, 1)$, $(2, 3)$, and $(3, y)$.

ANSWER:

The solution to the system of equations

$$\begin{aligned} a_0 &= 0 \\ a_2 + a_1 + a_0 &= 1 \\ 4a_2 + 2a_1 + a_0 &= 3 \end{aligned}$$

is $a_0 = 0$ and $a_1 = a_2 = \frac{1}{2}$. With these values, the graph of f passes through $(0, 0)$, $(1, 1)$, and $(2, 3)$. We can thus choose

$$y = f(3) = \frac{1}{2}(3^2 + 3) = 6.$$